

Bound clock: evidence and limits

Signal Space / GROSS Test 6

Technical completion: completed; bounded scientific classification: pass.

Run run-080a63d117abd84d; analysis analysis-0002-04d7ed8c.

Solver revision 8c42249f03b447e814c9d8017ce2cc02c0484eed; renderer revision 1c401dc6450094d3d88cd933eb42f6cb740e6d03.

Flat decoupling limit of SS OCF 1, $c=hbar=m=1$; constant internal doublet direction, $a=0$, gravity decoupled.

Four trial core frequencies 0.868, 0.875, 0.90, 0.94. Independently solved bound spectrum and trapping-off control.

Nominal peak $\chi=0.001$ for 100 eigenperiods; zero, half and double controls for 20 periods; independent $h/2$ and $2R$ controls for 20 periods.

Selected branch: 0.900. Numerically unresolved trial profiles: 0.875, 0.868; these are not evidence against those branches.

Outer energy sink is recorded, but background energy residual is too large relative to clock-mode energy to isolate radiated clock energy.

No emergent geometry, Test 7 reception or nonspherical stability established.

core-and-well

Question: Which trial core supplies a spatial mass well and localized clock eigenfunction?

Reading: Selected $\omega_Q=0.900$; core center $F=0.7426$; clock eigenvalue 0.170362 below vacuum 0.25.

Significance: The declared core density produces a spatially resolved clock mass well and a separately solved localized mode.

Limitation: Flat spherical branch only; neither gravity nor nonspherical perturbations follow from this profile.

bound-control

Question: Does the bound eigenvalue survive resolution controls and disappear without trapping?

Reading: Selected $E/Q=0.919412$; removing ν gives lowest eigenvalue 0.251711 versus 0.25 threshold.

Eigen error estimate $2.82e-07$.

Significance: The trapping coefficient is causally relevant to this bound mode in the registered family.

Limitation: A finite-box eigenvalue above 0.25 is a nonbound control. Radial Hessian diagnostics do not exclude all nonlinear or angular instabilities.

local-trace

Question: Does a local field trace tick for 100 periods when the unexcited core does not?

Reading: Nominal local trace completes 100.00 inferred signed cycles; local frequency 0.412749 versus eigenfrequency 0.41275. Unexcited trace is zero.

Significance: Unlike the stationary core density/projector, the field sampled inside the core has a repeatable local zero-crossing record.

Limitation: This is an ideal local field probe in the flat limit. Marker coupling, device noise and invariant reception comparison are future tests.

Bound clock: evidence and limits

lifetime-convergence

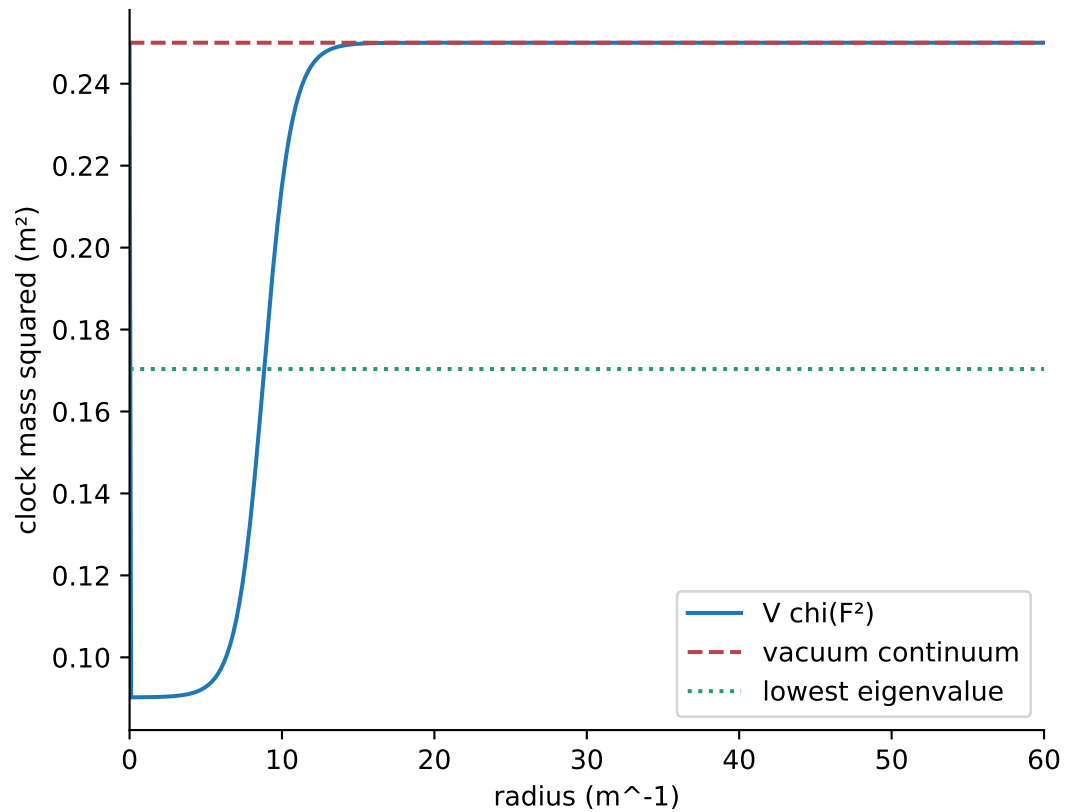
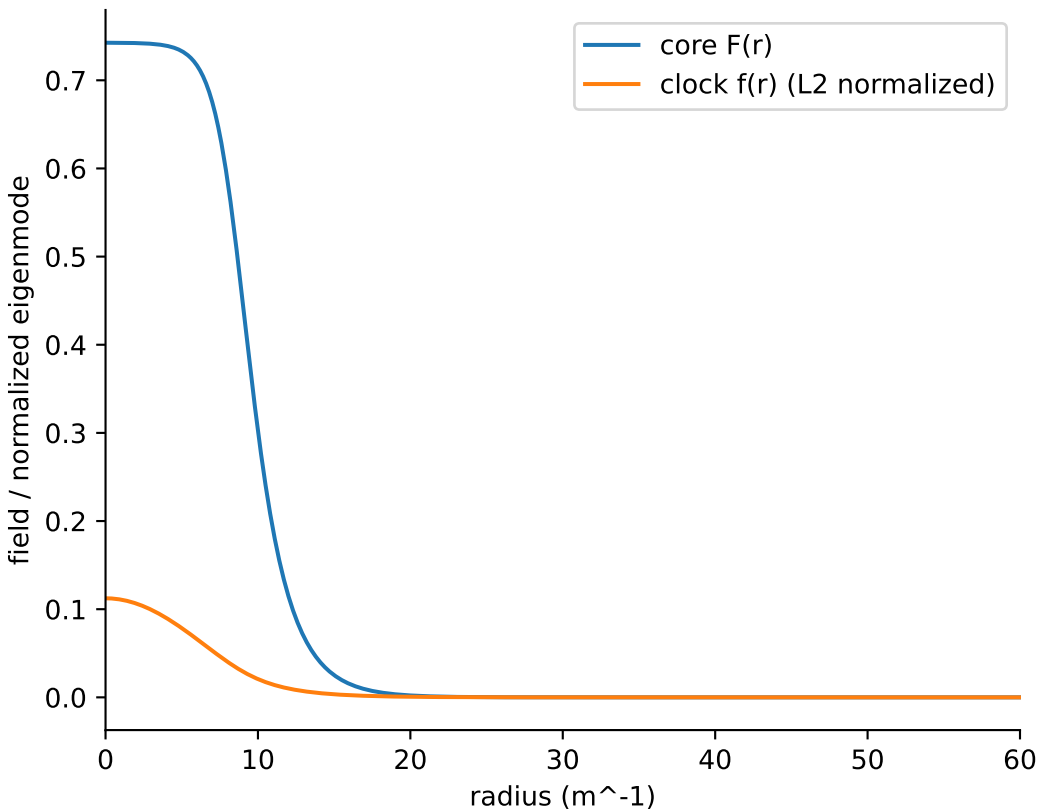
Question: Does the mode persist with controlled charge, amplitude, grid and boundary errors?

Reading: Over 100 periods mode-energy loss $2.53\text{e-}05$; maximum charge ledger error $1.44\text{e-}15$; short-run frequency grid/domain difference $1.01\text{e-}05$. Total-energy ledger residual is 13.4 initial clock-mode energies.

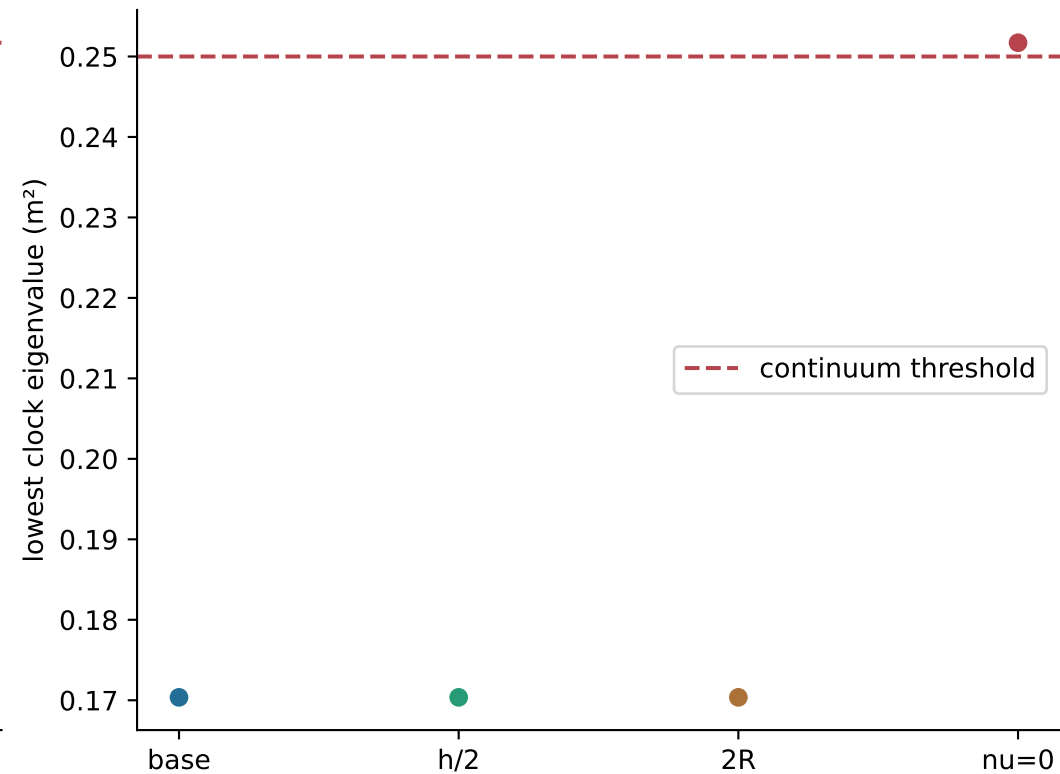
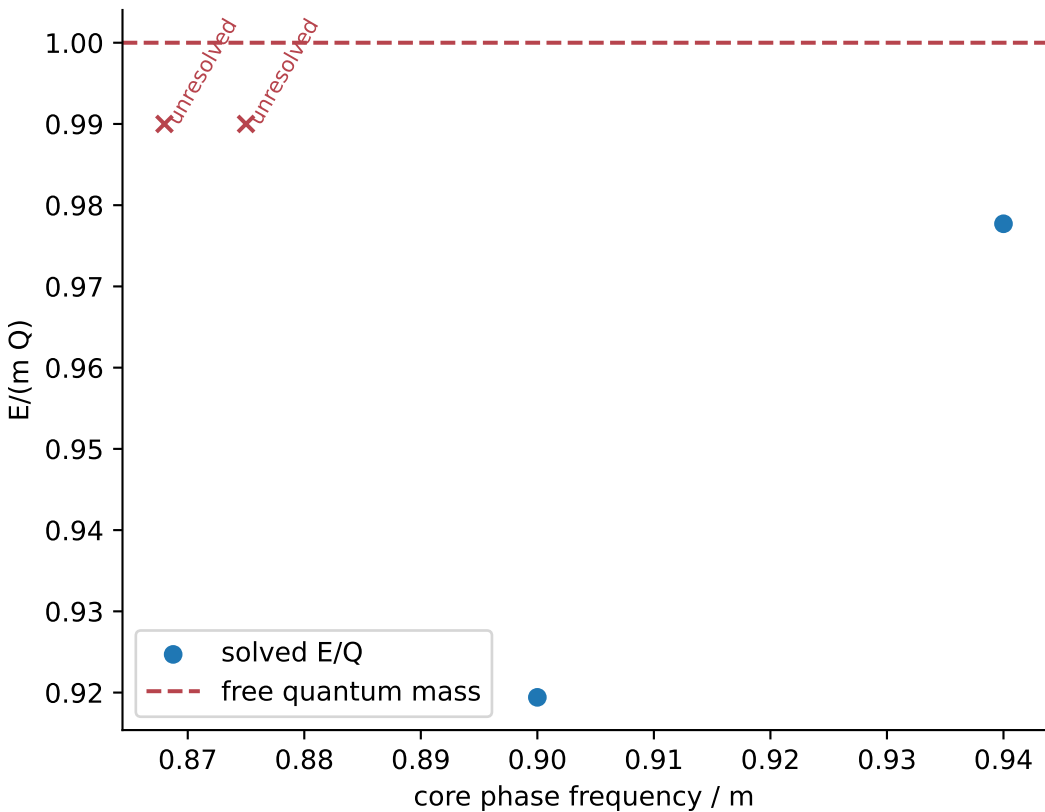
Significance: The duration and numerical controls quantify whether this core can provisionally serve as Test 7 calibration.

Limitation: Refined/wider runs and half/double amplitude controls cover 20 periods. Background-scale energy error exceeds the clock energy, so sponge loss cannot be assigned to clock radiation; nonspherical dynamics are untested.

Spatial core and independently computed clock mode

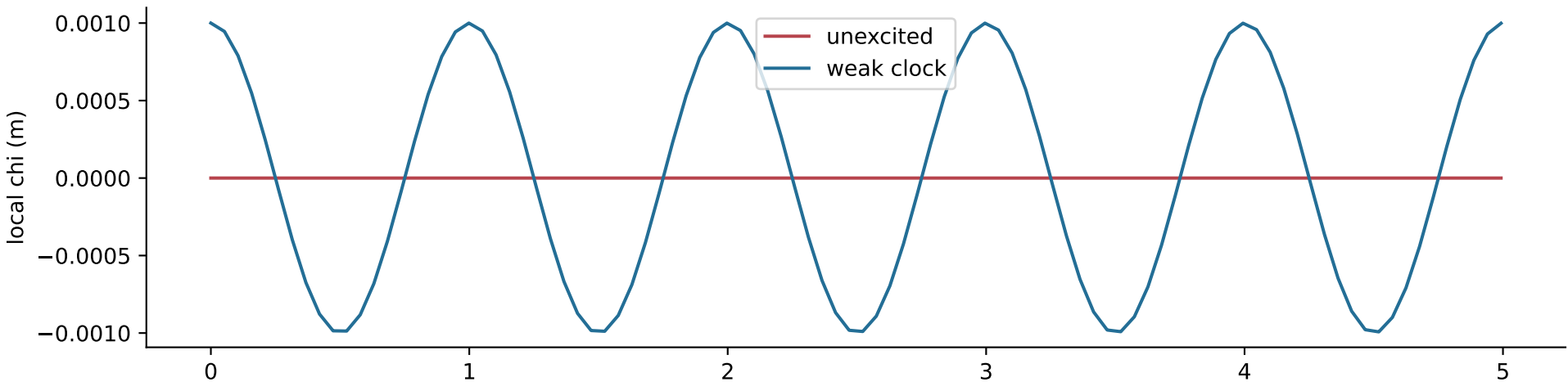


Core energetics and trapping failure control



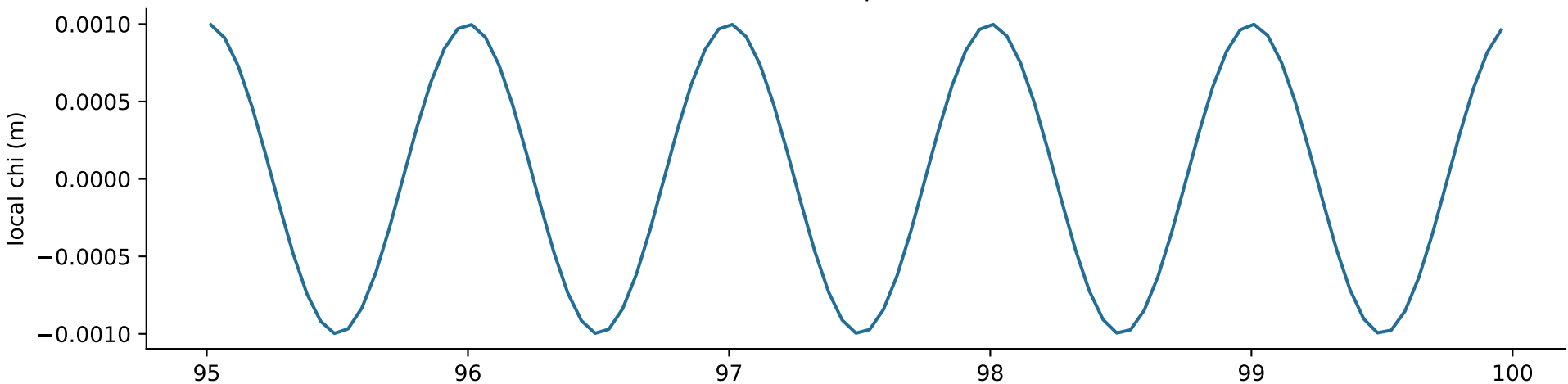
Local readout near core center, not a global fitted projection

first five periods



time / eigenperiod

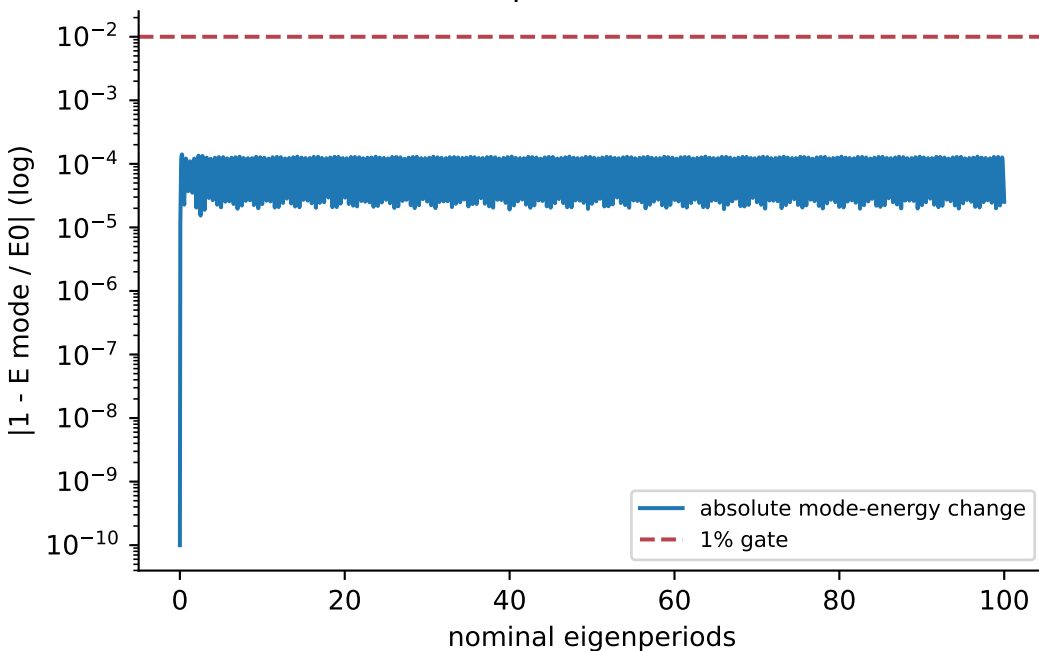
last five periods



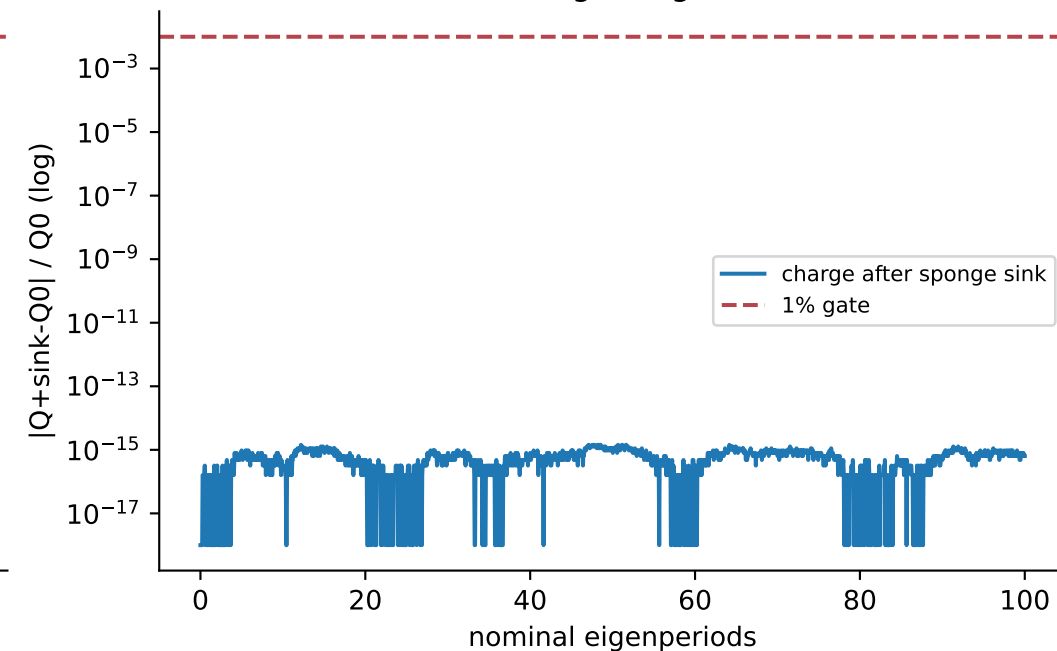
time / eigenperiod

Mode, charge and background energy ledgers with frequency controls

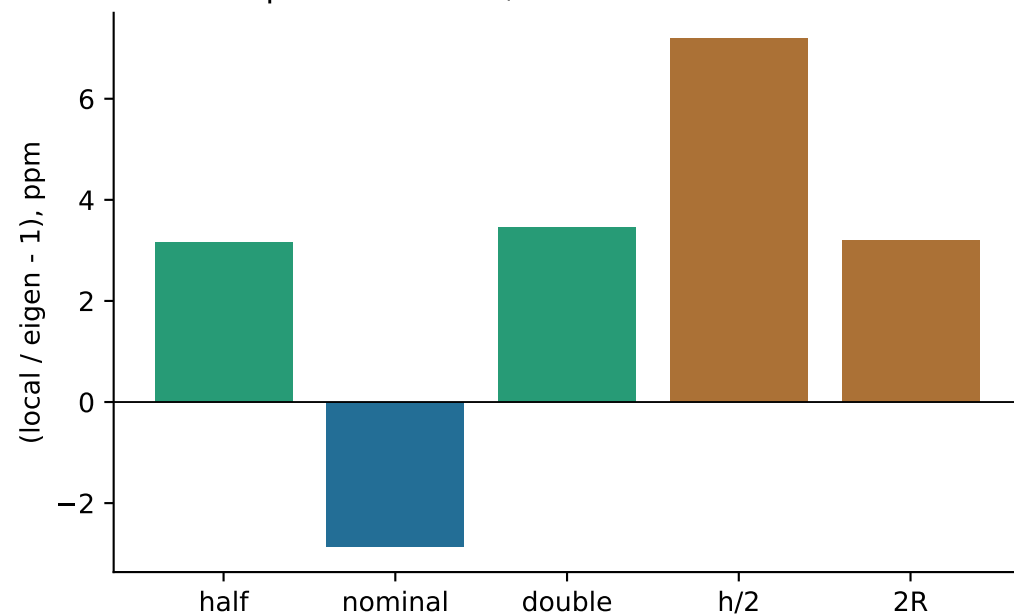
100-period lifetime



charge ledger



20-period controls; nominal shown over 100



Clock-scale radiation energy unresolved

